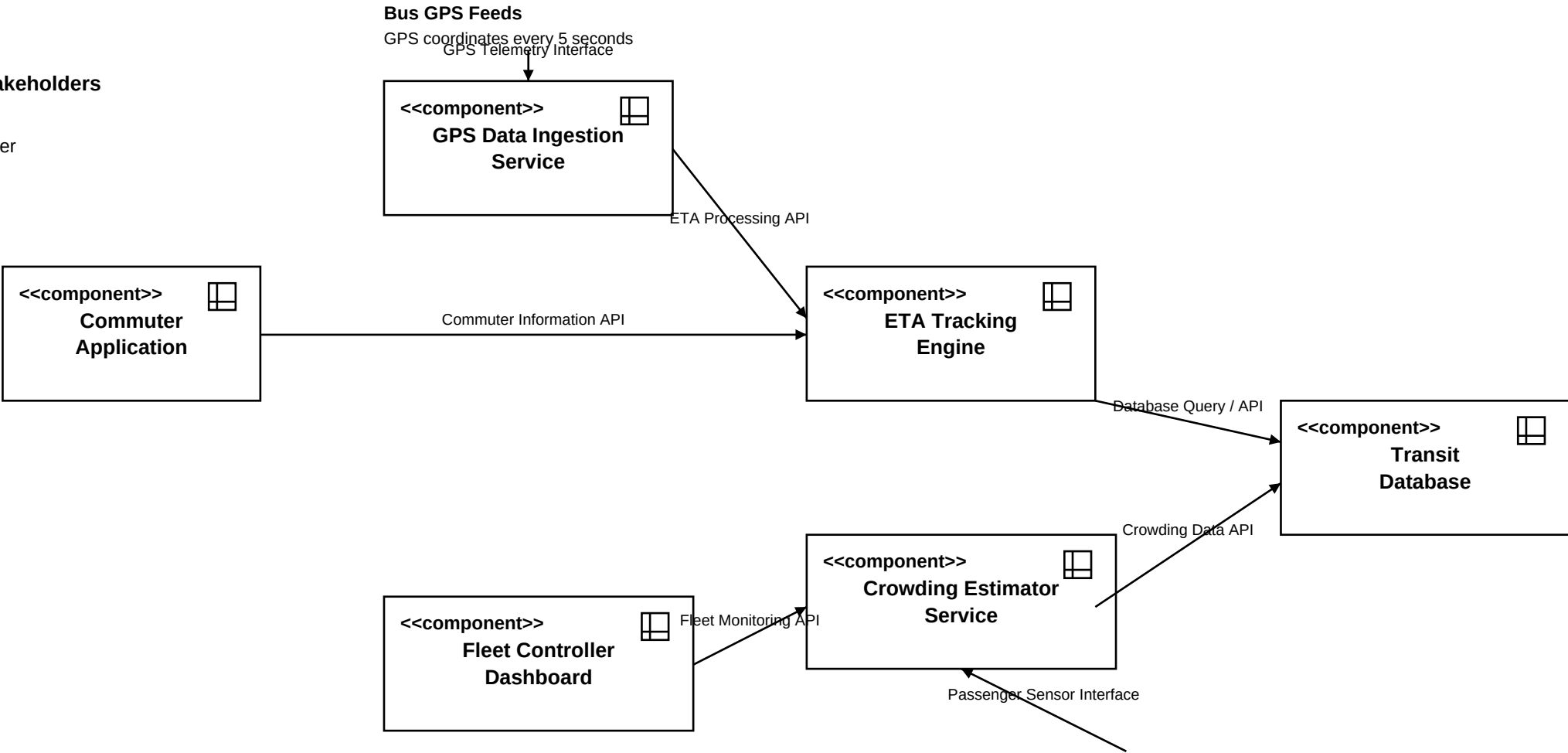


Public Bus Live Tracking & Crowding Estimator

UML Component Diagram — Microservices Architecture

Actors / Stakeholders

Commuter
Fleet Controller



FR-001 — GPS & ETA (Priority: High)

The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.

Pass: ETA updates based on current traffic speed and distance.

Fail: ETA increases while bus approaches stop.

NFR-001 — Performance & Security

The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.

Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Ticketing Sensors

Passenger / ticketing data